Solution:

If B does the work in 3 days, then A will do it in 1 day. That is, the difference is 2 days. Here, given that the difference between A and B in completing the work is 24 days. Therefore, A will take $\frac{24}{2} = 12$ days and B will take $3 \times 12 = 36$ days to complete the work separately.

Hence, the time taken by A and B together to complete the work = $\frac{ab}{a+b}$ days.

$$= \frac{12 \times 36}{12 + 36} = \frac{12 \times 36}{48} = 9 \text{ days.}$$

Note

If A is $\frac{a}{b}$ times as good a worker as B, then A will only $\frac{b}{a}$ of the time taken by B to complete the work. Try to solve the above Example 4.25 using this.

4.6.1 Sharing of the money for work

When a group of people do some work together, based on their individual work they get a share of money themselves. In general, **money earned** is shared by people, who worked together, in the ratio of the **total work** done by each of them.

DO YOU KNOW?

- If the ratio of the time taken by A and B in doing a work is $x : y$, then the ratio of work done by A and B is $\frac{1}{x} : \frac{1}{y} = y : x$. This is the ratio for their separate wages too.
- If three persons A, B and C can do a work in x, y and z days respectively, then the ratio in which their wages will be distributed to them is $\frac{1}{x} : \frac{1}{y} : \frac{1}{z}$.

Example 4.26



X, Y and Z can do a piece of job in 4, 6 and 10 days respectively. If X,Y and Z work together to complete, then find their separate shares if they will be paid ₹ 31000 for completing the job.

Solution:

Since they all work together for the same number of days $\left(\frac{60}{31}\right)$, the ratio in which they share the money is equal to the ratio of their work done per day.

That is, it is equal to $\frac{1}{4} : \frac{1}{6} : \frac{1}{10} = \frac{15}{60} : \frac{10}{60} : \frac{6}{60} = 15 : 10 : 6$

Here, the total parts = 15 + 10 + 6 = 31

Hence, A's share = $\frac{15}{31} \times 31000 = ₹15000$, B's share = $\frac{10}{31} \times 31000 = ₹10000$ and

C's share is ₹31000 − (₹15000 + ₹10000)= ₹6000.

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